OmniMed-FL: A Controlled Multimodal Federated Learning Study for Clinical Classification

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***Abstract*—We present OmniMed-FL, a controlled multimodal federated-learning study for five-class image–text classification. The proxy corpus contains 2,400 public chest radiographs, 600 synthetic COVID-19 image surrogates, and 3,000 class-conditioned synthetic notes; modalities are paired by class rather than by patient. Public DistilBERT and ViT-Base/16 encoders are fine-tuned from random task heads without pooled-data pretraining in the operational protocol. We vary Dirichlet heterogeneity, client count, anti-collapse components, initialization, eight fusion rules in a pooled-data diagnostic screen, and the federated update rule. Runtime and peak allocated GPU memory are observed, whereas FP32 communication is calculated from trainable parameters. Across two seeds, FedAvg Macro F1 changes from 0.896 ± 0.004 at α=5 to 0.674 ± 0.185 at α=0.1, exposing substantial sensitivity to the realized client partition. Because the text is templated, one image class is synthetic, and no patient-level pairing is available, these are proxy systems comparisons rather than estimates of clinical performance, privacy, or deployment readiness. At α=0.1, the matched results of Table V are local-only 0.310 (five-client mean), FedAvg 0.652 ± 0.057, FedProx 0.685 ± 0.064, and SCAFFOLD-style AdamW 0.192 ± 0.129. Thus FedAvg and FedProx outperform isolated local training, while the AdamW control-variate adaptation is unstable; the 0.033 FedProx–FedAvg gap is descriptive rather than statistically resolved.**

***Index Terms*—Multimodal Federated Learning, Non-IID Data, Vision-Language Models, Clinical Classification, Communication Cost, Retrieval**

I. Introduction

Clinical prediction often draws on both images and textual context. Centralized image models [1], [2] and medical vision–language models can combine these signals, but cross-institutional training would ordinarily require data to leave its source. Federated learning (FL) instead exchanges parameter updates while retaining each client's raw examples [3]. This data-locality property is useful, although FL alone is not a privacy guarantee and does not remove risks from model updates.

Federated medical imaging [4]–[6] and centralized medical vision–language modeling have mature precedents, with recent work also studying their intersection. The combination still raises questions that a single favorable run cannot answer: how performance changes with client skew, whether anti-collapse components help, whether fusion rankings survive seed variation, and what additional memory and communication the multimodal branch incurs.

We study these questions through OmniMed-FL, a controlled five-class proxy benchmark. Its public radiographs and synthetic clinical-style notes are not patient-matched data, so the work evaluates the learning system rather than clinical diagnosis. This scope permits reproducible matched comparisons while making the boundary between a systems result and a clinical claim explicit.

A. Contribution

The main *contributions* of this paper are as follows:

- We implement a reproducible multimodal FL protocol that begins from public pretrained encoders and random task heads, never from a model trained on pooled client examples.
- We evaluate data heterogeneity, client count, anti-collapse components, initialization, eight fusion rules, and matched FedAvg/FedProx/SCAFFOLD-style baselines with repeated seeds.
- We report realized client skew, wall-clock time, peak allocated GPU memory and FP32 communication payload, and evaluate retrieval over the full held-out split rather than a five-query demonstration.

II. Related Work

A. Medical Image Classification

Early work on automated diagnosis mainly treated radiology as an image classification problem. Rajpurkar *et al.* [1] and Irvin *et al.* [2] showed the effectiveness of CNN-based models for chest X-ray classification in centralized settings. Later work explored Vision Transformers [7], [8], which capture broader spatial relationships within an image. Although these approaches achieve strong classification performance, they generally rely on centrally collected datasets and do not make use of the clinical notes that often accompany medical images.

B. Federated Clinical Learning

Federated Learning (FL) has been studied as a way to train clinical models without moving patient data outside

TABLE I: Positioning vs. Related Work.

	Modality		Framework	
Work	**Img**	**Text**	**FL**	**VLM**
CheXNet [1]	✓	×	×	×
Sheller 2020 [4]	✓	×	✓	×
Kaissis 2021 [6]	✓	×	✓	×
BiomedCLIP [9]	✓	✓	×	✓
MedCLIP [11]	✓	✓	×	✓
FedMME [20]	✓	✓	✓	✓
OmniMed-FL (ours)	✓	✓	✓	✓

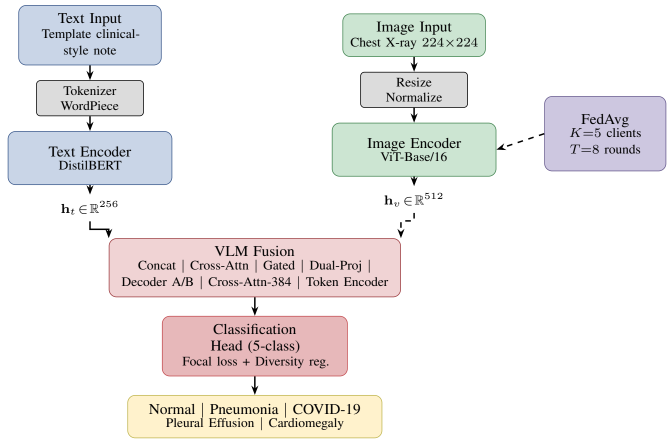

Fig. 1: OmniMed-FL architecture. DistilBERT and ViT-Base/16 encode the two proxy modalities; VLM fusion combines them; FedAvg coordinates simulated clients without transmitting their raw examples.

individual institutions. Sheller *et al.* [4] applied FedAvg in a decentralized medical imaging setting and Dou *et al.* [5] validated a federated model across national sites, while Kaissis *et al.* [6] considered additional privacy protection through differential privacy. FedProx [18] and SCAFFOLD [12] address optimization under heterogeneous clients through a proximal objective and control variates, respectively. Most clinical studies nevertheless remain image-only, motivating our matched multimodal comparison.

C. Vision-Language Models in Healthcare

Another line of work has focused on combining medical images with textual information. Models such as BiomedCLIP [9], LLaVA-Med [10], and MedCLIP [11] learn relationships between medical images and associated reports or clinical text. This allows the model to use information from both modalities rather than treating the image independently. However, these models are generally developed and trained in centralized settings, which makes their direct use across hospitals with isolated patient data difficult.

Synthesis: The intersection is no longer empty. FedMME uses a vision-language model to generate reports for a one-shot federated ensemble [20], while P-FIN studies uncertainty-aware imputation and aggregation when clinical sites have missing modalities [21]. These systems address different problems and data regimes, so cross-paper scores are not comparable. Our narrower contribution is a controlled multi-round study of image–text fusion, partition heterogeneity, anti-collapse components, and matched update rules; we make no priority or state-of-the-art claim.

III. System Design

A. Architecture

Figure 1 shows the OmniMed-FL pipeline. Clinical-style notes are tokenized with WordPiece to 128 tokens; chest images are resized to 224×224 and ImageNet-normalized. DistilBERT's `[CLS]` representation is projected to $\mathbf{h}_t \in \mathbb{R}^{256}$, while the mean-pooled ViT token sequence is projected to $\mathbf{h}_v \in \mathbb{R}^{512}$.

The two representations are combined by eight lightweight operators: 384-dimensional concatenation, residual cross-attention, a gated residual, 256-dimensional dual-projection concatenation, two registry variants with the same decoder-layer parameterization, non-residual 384-dimensional cross-attention, and a two-token Transformer encoder. Historical code keys refer to several vision–language model families, but these operators are not implementations of those foundation models. Each produces $\mathbf{h}_f$, which is passed to a two-layer five-class MLP classifier.

We include simple and more complex fusion rules to test whether additional attention structure produces a repeatable benefit under the same training budget. The repeated-seed comparison is a pooled-data diagnostic architecture screen reported in Section IV, not an operational FL result; we do not infer an ordering from the earlier single-run screening plot.

B. Federated and Local Objectives

Let K clients hold disjoint shards $\mathcal{D}_k$ of the training split, with $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$, and let θ collect the trainable parameters on the fusion path: both encoders, the projection layers, the fusion operator and the classification head. The federated objective is the sample-weighted empirical risk

$$\min_{\theta} \; F(\theta) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(\theta), \quad F_k(\theta) = \frac{1}{n_k} \sum_{(x,z,y)\in\mathcal{D}_k} \ell\big(f_\theta(x,z), y\big), \tag{1}$$

where x is a radiograph, z its class-paired note and f_θ the fused five-class classifier. Shards are never exchanged; the server receives only the trainable tensors enumerated above.

The local loss combines a focal term with a batch-level diversity penalty,

$$\ell = (1-p_y)^\gamma \, \mathrm{CE}_\epsilon\big(f_\theta(x,z), y\big) \;+\; \lambda\Big(1 - \tfrac{H(\bar{p})}{\log C}\Big), \tag{2}$$

with focusing parameter γ=2, label smoothing ϵ=0.05, C=5 classes, and $\bar{p}$ the mean predicted distribution over the mini-batch. The second term grows when a client's predictions concentrate on few classes, which is the failure mode a severe Dirichlet split invites. Setting λ=0 removes it; together with a class-balanced batch sampler it forms the anti-collapse stack ablated in Section IV.

Algorithm 1 OmniMed-FL federated training

Require: clients K, rounds T, local epochs E, diversity weight λ

initialize $\theta^{(0)}$ from public DistilBERT and ViT-Base/16 weights, with randomly initialized projection, fusion and classification heads
for $t = 0$ **to** $T - 1$ **do**
  for each client $k = 1, \ldots, K$ in parallel **do**
    $\theta_k \leftarrow \theta^{(t)}$
    for $e = 1$ **to** E **do**
      for each class-balanced mini-batch $B \subset \mathcal{D}_k$ **do**
        $g \leftarrow \nabla_{\theta_k} \ell(B)$ by Eq. (2)
        clip $\|g\|_2 \leq 1$; $\theta_k \leftarrow \mathrm{AdamW}(\theta_k, g)$
      end for
    end for
  end for
  $\theta^{(t+1)} \leftarrow \sum_{k=1}^{K} \frac{n_k}{n} \theta_k$ (trainable tensors only)
end for
return $\theta^{(T)}$

Algorithm 1 states the protocol as executed. The initialization line is the operational constraint that the audit added: every reported federated run starts from public encoder weights and random task heads, so no pooled-data model can leak into a federated result.

C. Encoding Specifications

The text branch uses the public DistilBERT checkpoint [17], with six transformer layers and a 768-dimensional hidden state. The image branch uses the public ViT-Base/16 checkpoint [7], with 12 transformer layers, a 768-dimensional hidden state and 16×16 input patches. Both pretrained encoders are fine-tuned locally. Projection layers map their representations to the fusion-specific latent size, and the operational federated protocol starts with these public encoder weights and randomly initialized projection, fusion and classification heads. No model trained on pooled client data initializes an operational FL run. The unimodal classifier heads and inactive CNN fallback are excluded from VLM optimization and communication because they are not on the fusion path.

D. Dataset and Clinical Corpus

Image data. We form a balanced 3,000-image proxy corpus (600 examples per class). The public radiographs come from the Kermany pneumonia collection [15] and the NIH ChestX-ray dataset [16]. Loader logs verify 2,400 public images covering Normal, Pneumonia, Pleural Effusion and Cardiomegaly. The named COVID-19 source was unavailable at execution time, so the 600 COVID-19 images are synthetic X-ray-style surrogates with bilateral opacity patterns. Images are resized to 224×224, normalized with ImageNet statistics, shuffled deterministically and split 80/20 into 2,400 training and 600 validation examples.

Clinical text data. No patient EHR data are used. Each image label is paired at the class level with a synthetic clinical-style note generated from observation, symptom, context and indicator slots. Twenty-five percent of notes draw all slots from the target class, 35% mix target and non-target slots, and 40% are heavily mixed while retaining a probabilistic target-class indicator. This controlled construction reduces direct label wording, but the image and note are not from the same patient and must not be described as a real image–report pair.

What this corpus does and does not support. Every experimental arm sees the same controlled corpus, so matched comparisons between federated algorithms, fusion rules and ablations are meaningful within this benchmark. The absolute scores are not diagnostic-performance estimates: one image class is synthetic, all text is templated, pairing is label-level, and the split is not patient-grouped across source institutions. We therefore make no claim about clinical utility, deployment readiness or generalization to real paired radiology reports.

TABLE II: Global Hyperparameter Configuration.

Parameter	Value
Local Epochs (E)	3
Federated Rounds (T)	8
Batch Size	16
AdamW Learning Rate	1×10^{-4}
Weight Decay	0.01
Diversity Reg. (λ)	1.0
Reference K, α	5, 1.0
Seeds	0, 1
Initialization	Public encoders + random heads
Precision	FP32 tensors (no AMP)

E. Experimental Setup

Hyperparameters. Each client uses AdamW with learning rate 10^{-4} and weight decay 0.01 for three local epochs per round. The reviewer suite uses eight rounds, batch size 16, FP32 tensors without automatic mixed precision, and diversity-loss weight 1.0. The reference setting is five clients with Dirichlet $\alpha = 1.0$; Section IV varies α and the number of clients explicitly. Seeds 0 and 1 jointly control task-head initialization and the realized client partition. Experiments run in PyTorch on NVIDIA H100 NVL GPUs shared with unrelated workloads; wall-clock measurements therefore include uncontrolled contention. CUDA deterministic algorithms are not forced, so the seeds control software RNG streams but do not guarantee bitwise-identical reruns. Table II records the fixed protocol.

Partitioning. For each class, a Dirichlet draw allocates training indices across clients. Smaller α creates more concentrated client label distributions; $\alpha \to \infty$ approaches IID allocation [13]. Because nominal α does not fully describe a finite realized split, we also report the mean client share occupied by its dominant class. The same split is held fixed across rounds for a given seed.

Rationale for Table II. The values are fixed across matched arms; only the factor named by an ablation or sweep changes. Three local epochs and eight rounds provide a 24-local-epoch budget while keeping the full sweep tractable. Batch size 16 holds both active encoders and a local model copy in memory. The 10^{-4} rate and diversity weight 1.0 are the implemented defaults rather than outcomes of per-model tuning. Consequently, we interpret comparisons within the grid and do not claim each configuration is individually optimal.

TABLE III: Effect of partition heterogeneity on Fed-VLM. α is the Dirichlet concentration; *skew* is the realized mean per-client share of the dominant class and diversity is the minimum predicted-class fraction across rounds. Mean ± std over seeds.

α	Skew	Macro F1	Min. div.
0.1	0.74	0.674 ± 0.185	0.90
0.3	0.68	0.769 ± 0.043	1.00
0.5	0.63	0.824 ± 0.113	1.00
1.0	0.48	0.837 ± 0.045	1.00
5.0	0.29	0.896 ± 0.004	1.00

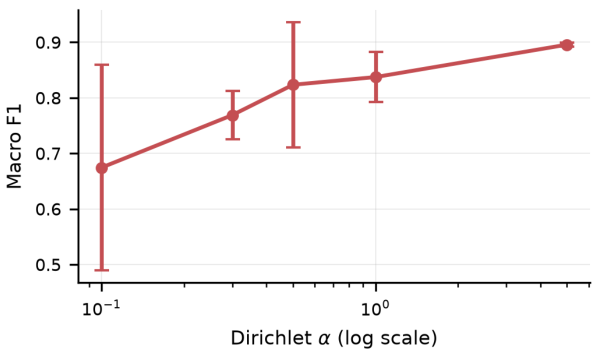

Fig. 2: Macro F1 against Dirichlet α (log scale). Markers are the two-seed mean, bars the sample standard deviation. The mean declines monotonically as the partition sharpens, but the across-seed spread widens much faster than the mean moves.

IV. Results

A. Heterogeneity and Client Scaling

Table III shows that the nominal Dirichlet concentration and the realized dominant-class share move together. Macro F1 falls from 0.896 ± 0.004 at $\alpha{=}5$ to 0.674 ± 0.185 at $\alpha{=}0.1$. The severe setting is therefore 0.163 below the reference $\alpha{=}1$ result and is substantially less stable across seeds. Minimum predicted-class diversity remains high, so the principal failure is degraded and variable discrimination rather than permanent single-class prediction.

Figure 2 plots the same sweep and makes the variance structure explicit. The seed spread grows from ± 0.004 at $\alpha{=}5$ to ± 0.185 at $\alpha{=}0.1$, where the two seeds straddle 0.489–0.859. This is the methodological point of the study in one picture: at severe skew the realized partition, rather than the update rule, dominates the outcome, and a single favorable run carries almost no information.

Table IV varies the client count at $\alpha{=}1.0$. Quality is not monotone in K: the ten-client setting is highest in this grid and $K{=}20$ is slightly lower. Mean wall-clock grows from 545 to 908 seconds, while calculated full-run communication grows linearly from 27.5 to 183.5 GiB, since the payload per round scales with the number of participating clients. Because the corpus size is fixed, larger K also means smaller shards, which confounds partition difficulty with client count; and because the runs share a contended server, none of these timings is a claim of dedicated multi-node speedup.

TABLE IV: Scalability in the number of clients K at $\alpha{=}1.0$. Communication is computed from the FP32 trainable-parameter payload, $V = 2KT|\theta|b$; wall-clock is observed on a contended shared server. Mean ± std over seeds.

K	Macro F1	Wall-clock (s)	Total comm. (GiB)
3	0.812 ± 0.043	545	27.5
5	0.825 ± 0.082	708	45.9
10	0.863 ± 0.008	893	91.8
20	0.846 ± 0.004	908	183.5

TABLE V: Federated algorithms under *matched* settings: same Concat-VLM backbone, data, partition, local budget and hyperparameters. Among federated rows, the update rule is the only variable; the local-only row is the no-federation reference. Federated rows are mean ± std over seeds.

Method	$\alpha = 0.1$
Local only (no federation)	0.310 [0.064, 0.575]
FedAvg	0.652 ± 0.057
FedProx	0.685 ± 0.064
SCAFFOLD-style (AdamW)	0.192 ± 0.129

Local-only is the mean over the five clients, each trained on its own shard alone, evaluated at its final matched-budget epoch; brackets give the client range for that seed-0 partition. Every other row is the global model after T rounds.

B. Matched Federated Baselines

Table V is the principal algorithm comparison: each federated arm uses the same Concat-VLM, partition, optimizer and 24-local-epoch cumulative budget. The local-only row averages the five client models at their final matched epoch; it does not select a favorable validation epoch. SCAFFOLD is reported as "SCAFFOLD-style (AdamW)" because these local steps are an adaptation rather than the classical SGD algorithm. FedAvg (0.652 ± 0.057) and FedProx (0.685 ± 0.064) exceed the local-only mean (0.310), although the 0.033 difference between them is smaller than their across-seed standard deviations. The SCAFFOLD-style adaptation reaches only 0.192 ± 0.129 and transiently predicts as few as one of five classes. This is evidence that the tested AdamW adaptation is unstable under the severe partition, not a result about classical SGD SCAFFOLD.

C. Components and Initialization

The component ablation in Table VI holds the optimizer-step budget fixed. At $\alpha{=}0.1$, removing the balanced sampler lowers F1 from 0.716 ± 0.014 to 0.585 ± 0.148, whereas removing only the diversity term gives 0.701 ± 0.011 and removing both gives 0.657 ± 0.057. Minimum predicted-class diversity declines from 0.80 to 0.60 when both are removed, but no arm remains in a single-class state. At $\alpha{=}1$, all four means lie between 0.809 and 0.834, so the stack has no resolved benefit in the milder partition. In the separate pooled-data arm, early abort improves centralized F1 from 0.845 ± 0.013 to 0.884 ± 0.003. The pooled-data initialization begins higher than the operational cold start (round-1 F1 0.843 versus 0.704) but finishes slightly lower (0.818 ± 0.004 versus 0.839 ± 0.065); it is therefore a non-deployable trajectory diagnostic, not an operational initialization or a final-performance advantage. The full-stack anchor is rerun inside the ablation block; because deterministic CUDA

TABLE VI: Anti-collapse component ablation on Fed-VLM. Each row removes one component from the full stack. Diversity is the minimum fraction of the five classes predicted in any round. Mean $\pm$ std over seeds.

	$\alpha = 0.1$		$\alpha = 1.0$	
Configuration	F1	Div.	F1	Div.
Full stack	0.716 ± 0.014	0.80	0.821 ± 0.043	1.00
$-$ balanced sampler	0.585 ± 0.148	0.70	0.809 ± 0.064	1.00
$-$ diversity term	0.701 ± 0.011	0.70	0.834 ± 0.043	1.00
$-$ both	0.657 ± 0.057	0.60	0.823 ± 0.073	0.90

TABLE VII: Initialization ablation. The pooled-data oracle is diagnostic only and is not part of the privacy-preserving FL protocol.

Initialization	Macro F1	Round-1 F1
Pooled-data oracle	0.818 ± 0.004	0.843
Public encoders + random head	0.839 ± 0.065	0.704

algorithms are not forced, it is used only for within-block deltas and is not spliced with the nominally identical E1 point.

Table VII isolates that initialization comparison. The pooled oracle buys a large head start—round-1 Macro F1 0.843 against 0.704—and then gives it back, finishing 0.021 below the operational cold start with a much narrower seed spread. Two readings are consistent with this: the cold start has not converged by round 8 in one seed, or the oracle has already fitted the pooled distribution and transfers no additional advantage under the federated update. Either way, the pooled-initialization result that the earlier draft reported as a federated retention figure is not reproduced once the initialization is made operational, which is why no headline retention number is claimed here.

D. Fusion, Cost, and Retrieval

Decoder-style B has the highest descriptive mean in Table VIII (0.913 ± 0.008), followed by the two-token encoder (0.902 ± 0.047) and decoder-style A (0.899 ± 0.037). The 0.754–0.913 range is wider than several within-operator seed spreads, but two seeds do not support a statistically resolved ranking among close means. Moreover, decoder-style A and B share the same parameterization, so their difference is execution variability rather than an architectural contrast. This pooled-data screen selects no initialization for operational FL; those runs begin from public encoders and random task heads.

The communication entries in Table IX follow

$$V = 2KT|\theta|b, \tag{3}$$

where $b{=}4$ bytes for FP32 trainable parameters. Payload is calculated, not observed on a network. Wall-clock time and peak allocated memory are measured, but the shared H100 server had uncontrolled competing workloads, so timing is a record of these executions rather than a dedicated-throughput benchmark.

Retrieval uses exact FAISS inner-product search [19] over TF–IDF vectors for every held-out query. Across all 600 queries (Table X), top-1 label accuracy is 0.522, precision@5 is 0.461, and mean top-1 cosine similarity is 0.707. The per-condition rows are tightly clustered—top-1 accuracy spans 0.484 to 0.545 and precision@5 spans 0.442 to 0.478—so no class is retrieved appreciably better than the others, and the similarity spreads (± 0.08 to ± 0.10) overlap completely across conditions. Against a five-class prior of 0.2 the index is clearly informative, but 0.522 top-1 accuracy is far from a reliable case-matching tool. This is an auxiliary label-retrieval test over synthetic notes: it is not generation, not a fine-tuned-DistilBERT retrieval result, and not clinical evidence. It replaces the five-query demonstration of the earlier draft, whose apparent near-perfect retrieval was an artifact of the probe size.

TABLE VIII: Pooled-data diagnostic screen of lightweight fusion operators with repeated seeds. This architecture screen is not an operational FL run. Registry variants A and B share the same decoder-layer parameterization and are not distinct foundation models. Values are the best validation Macro F1 within the fixed 12-epoch budget. The standard deviation quantifies seed sensitivity; with two seeds, close means are treated as descriptive rather than a statistically resolved ranking.

Fusion	Macro F1	Seeds
Decoder-style B (256-d)	0.913 ± 0.008	2
Two-token encoder (256-d)	0.902 ± 0.047	2
Decoder-style A (256-d)	0.899 ± 0.037	2
Residual cross-attn (256-d)	0.888 ± 0.012	2
Concat (384-d)	0.867 ± 0.044	2
Gated residual (256-d)	0.862 ± 0.021	2
Dual-projection concat (256-d)	0.841 ± 0.002	2
Cross-attn (384-d)	0.754 ± 0.015	2

TABLE IX: Cost per branch at $K{=}5$, $\alpha{=}1.0$. FP32 payload is computed from trainable parameters; wall-clock and peak allocated memory are observed on the shared GPU named in the results metadata and include uncontrolled contention.

Branch	$\lvert\theta\rvert$	Payload	Total V	s/round	Peak mem.
Fed-LLM	66.6 M	254 MiB	19.8 GiB	139	1731 MiB
Fed-ViT	86.9 M	332 MiB	25.9 GiB	252	3253 MiB
Fed-VLM	153.9 M	587 MiB	45.9 GiB	337	4884 MiB

V. Discussion

The controlled results isolate several systems effects while also exposing the limits of favorable single-run reporting. Partition severity changes both mean performance and seed sensitivity, so nominal α should be accompanied by the realized client histograms. Likewise, the repeated fusion and optimizer comparisons are more informative than rankings assembled from unmatched corpora.

Security and governance. The simulation keeps raw proxy examples in client datasets and aggregates model states, but it implements no network transport, secure aggregation, differential privacy, attack evaluation, or privacy accounting. FL data locality is therefore not a privacy guarantee; model updates remain functions of local examples. A deployment would require a threat model and mechanisms such as secure aggregation and, where appropriate, formally accounted differential privacy [14].

TABLE X: Retrieval evaluation over 600 held-out queries against a 2400-note index (top-k=5).

Condition	n	Top-1 acc.	P@k	Top-1 sim.
Normal	115	0.539	0.461	0.695 ± 0.096
Pneumonia	121	0.512	0.463	0.716 ± 0.098
Covid19	128	0.531	0.442	0.715 ± 0.090
Pleural Effusion	126	0.484	0.465	0.705 ± 0.090
Cardiomegaly	110	0.545	0.478	0.703 ± 0.079
Overall	600	**0.522**	–	0.707

Limitations. The 3,000 notes are synthetic, the 600 COVID-19 images are synthetic surrogates, and image–text pairing is by class rather than patient. The split is not grouped by patient or source institution, the task is single-label, and no external or prospective clinical validation is performed. Only two seeds are available and CUDA deterministic algorithms were not forced. The simulated clients participate in every round on one shared server; stragglers, dropout, malicious clients, network latency, and hardware heterogeneity remain untested. These restrictions make the relative comparisons internally useful, but the absolute scores are not estimates of diagnostic performance or deployment readiness.

VI. Conclusion

OmniMed-FL provides an audited controlled study of multimodal FL rather than a clinical system claim. Clean experiments span heterogeneity, client scale, anti-collapse components, initialization, fusion, matched update rules, systems cost, and held-out retrieval. The central finding is that conclusions depend materially on partition realization and comparison design; matched repeated runs replace the earlier pooled-initialization and cross-corpus claims. Evaluation on real patient-matched multi-institutional data, with privacy mechanisms and a deployment threat model, is required before drawing clinical conclusions.

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References

[1] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya, M. P. Lungren, and A. Y. Ng, "CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning," *arXiv:1711.05225*, 2017.

[2] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, et al., "CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison," in *Proc. AAAI Conf. Artificial Intelligence*, 2019, pp. 590–597.

[3] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.

[4] M. J. Sheller, B. Edwards, G. A. Reina, J. Martin, S. Pati, A. Kotrotsou, M. Milchenko, et al., "Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data," *Scientific Reports*, vol. 10, art. 12598, 2020.

[5] Q. Dou, T. Y. So, M. Jiang, Q. Liu, V. Vardhanabhuti, G. Kaissis, Z. Li, et al., "Federated deep learning for detecting COVID-19 lung abnormalities in CT: a privacy-preserving multinational validation study," *npj Digital Medicine*, vol. 4, art. 60, 2021.

[6] G. Kaissis, A. Ziller, J. Passerat-Palmbach, T. Ryffel, D. Usynin, A. Trask, et al., "End-to-end privacy preserving deep learning on multi-institutional medical imaging," *Nature Machine Intelligence*, vol. 3, pp. 473–484, 2021.

[7] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.

[8] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 10012–10022.

[9] S. Zhang, Y. Xu, N. Usuyama, H. Xu, J. Bagga, R. Tinn, S. Preston, et al., "BiomedCLIP: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs," *arXiv:2303.00915*, 2023.

[10] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, and J. Gao, "LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

[11] Z. Wang, Z. Wu, D. Agarwal, and J. Sun, "MedCLIP: Contrastive learning from unpaired medical images and text," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2022, pp. 3876–3887.

[12] S. P. Karimireddy, S. Kale, M. Mohri, S. J. Reddi, S. U. Stich, and A. T. Suresh, "SCAFFOLD: Stochastic controlled averaging for federated learning," in *Proc. Int. Conf. Machine Learning (ICML)*, 2020, pp. 5132–5143.

[13] T.-M. H. Hsu, H. Qi, and M. Brown, "Measuring the effects of non-identical data distribution for federated visual classification," *arXiv:1909.06335*, 2019.

[14] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in *Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS)*, 2016, pp. 308–318.

[15] D. S. Kermany, M. Goldbaum, W. Cai, C. C. S. Valentim, H. Liang, S. L. Baxter, et al., "Identifying medical diagnoses and treatable diseases by image-based deep learning," *Cell*, vol. 172, no. 5, pp. 1122–1131.e9, 2018.

[16] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, "ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2097–2106.

[17] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," *arXiv:1910.01108*, 2019.

[18] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst. (MLSys)*, vol. 2, 2020.

[19] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2019.

[20] N. Wang, Y. Deng, S. Fan, J. Yin, and S.-K. Ng, "Multi-modal one-shot federated ensemble learning for medical data with vision large language model," *arXiv:2501.03292*, 2025.

[21] N. F. Shahid, M. Ahmed, M. A. Haider, S. R. Sagor, A. Rahman, and M. A. Hossain, "Probabilistic feature imputation and uncertainty-aware multimodal federated aggregation," in *Proc. Medical Imaging with Deep Learning (MIDL)*, PMLR, vol. 315, pp. 3593–3607, 2026.